

Pantheon Lab: Creating hyper- realistic AI digital humans with powerful GPUs

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ABOUT PANTHEON LAB

By using the high performance GPU on Google Cloud to train models, Pantheon Lab is able to provide highly realistic digital humans rendered in real time for various purposes such as avatars in videos and customer service.



Pantheon Lab
POWERED BY IMAGINATION

About Pantheon Lab

Founded in 2019, Pantheon Lab is a tech company based in Hong Kong providing hyper-realistic AI-powered digital humans to automate video production and customer services.

Offering customized chatbot solutions and a SaaS platform that accelerates video creation with digital humans, Pantheon Lab currently serves thousands of enterprise customers in East Asia, Southeast Asia, and the US across diverse industries, including real property, public sector, financing, and education.

Google Cloud results

- Enables smooth image rendering for digital humans with 300% more VRAM offered by Cloud GPUs
- Supports stable ML model testing environments with GPUs running in Compute Engine
- Realizes cost-effective and secure data storage through Cloud Storage

Support faster training

Industries: Technology

Location: Hong Kong

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>)

Cloud GPUs (<https://cloud.google.com/gpu/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

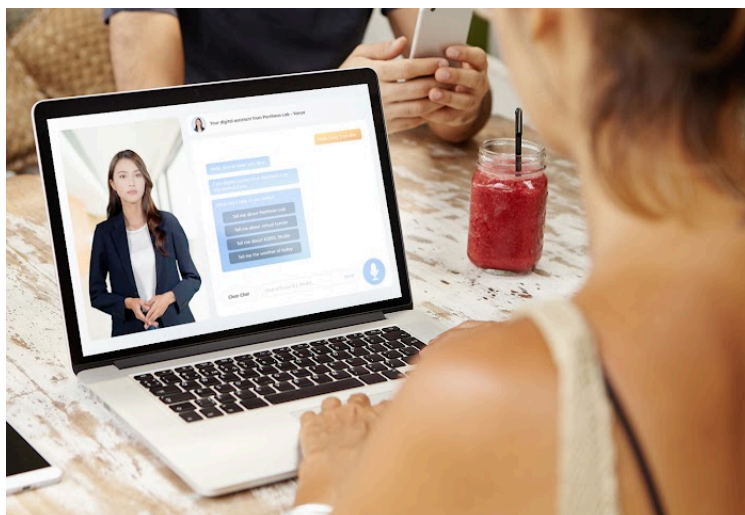
Various artificial intelligence (AI) technologies are being increasingly applied across industries to boost

automation, and digital humans are one of them. By employing digital avatars that deliver highly realistic body movements and facial expressions, companies can create video content that offers more engaging and efficient customer service.

Specialized in deep learning, [Pantheon Lab](https://www.pantheonlab.ai/)

(<https://www.pantheonlab.ai/>) is committed to building easy-to-adopt AI-powered digital humans for a wide range of use-cases. Founded in 2019, the Hong Kong-based company launched [AIDOL Studio](https://www.aidol.studio/)

(<https://www.aidol.studio/>), a one-stop SaaS platform that enables users to swiftly create digital avatars to be incorporated into their videos. Besides SaaS services, Pantheon Lab also develops customized digital human solutions, serving the public services sector, fast food chains, and retail businesses. It currently serves thousands of enterprise customers in East Asia, Southeast Asia, and the US across different industries, including real property, public sector, financing, and education.



"We're
constantly
developing and
testing new AI-
powered
products and
features, so
having abundant
computing
resources is
crucial to our
business.
Google Cloud
offers diverse
compute
machine types
and high
scalability that
can support our
resource-
intensive ML

compute workloads."

—*Ivan Lau, Co-Founder &
CEO, Pantheon Lab*

"With our cutting-edge digital human technology, we want to help enterprises in different sectors provide better services through a higher level of automation," says Ivan Lau, Co-founder & CEO at Pantheon Lab.

"Since we develop the technology on our own, we can flexibly adapt our solutions to customers' needs in various usage scenarios. Our customized chatbots support more than 50 languages, and have a rather high comprehension accuracy rate."

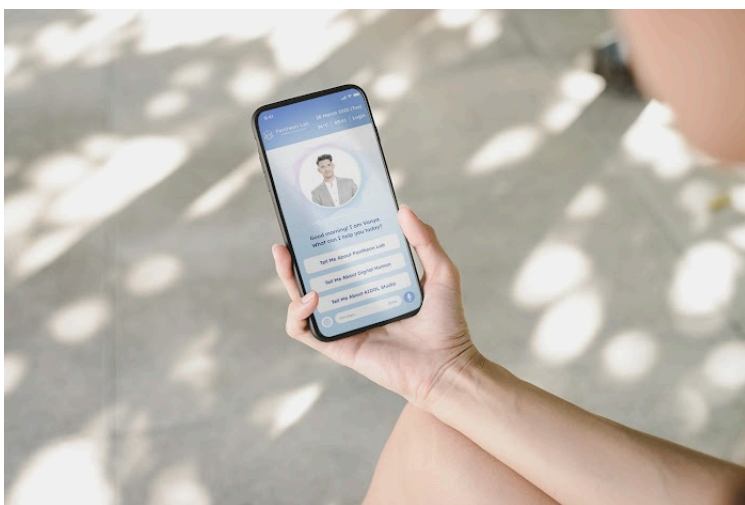
To develop digital avatars for diverse purposes and continue upgrading its products, Pantheon Lab needs to train the machine learning (ML) models that support its digital humans with a large amount of data. At first, the company used on-prem servers to run its ML models. However, as the business grew, the team realized that its existing on-prem computing resources couldn't be scaled up as fast as required. Pantheon Lab then decided to train its ML models in an environment that can offer more scalable computing resources and landed on Google Cloud (<https://cloud.google.com/>) in 2020 because of its high-performance and diverse computing products.

"We're constantly developing and testing new AI-powered products and features, so having abundant

computing resources is crucial to our business," explains Lau. "Google Cloud offers diverse compute machine types and high scalability that can support our resource-intensive ML compute workloads."

Enabling efficient and smooth ML model training for digital humans

Pantheon Lab now leverages Cloud GPUs (<https://cloud.google.com/gpu>) to train its ML models as a part of its research and development work, which largely consists of improving the image rendering of digital humans in real time to make them look more real people. Lau says that the A100 GPU model of Cloud GPUs has significantly facilitated the company's graphics-intensive workloads, because it delivers high performance with 300% more video random-access memory (VRAM), which is used to store graphics data for rendering, compared to consumer-grade GPUs.



Digital Assistant by Pantheon Lab

"Computing performance is key to the success of our research and development work. Cloud GPUs provide us with industry-leading GPU models that can be flexibly deployed to meet our needs for ML model training, which leads to higher efficiency and smoother development process."

—Ivan Lau, Co-Founder &
CEO, Pantheon Lab

At the same time, the diverse ways to run GPU instances on Google Cloud enables Pantheon Lab to flexibly arrange its instance deployment to meet its needs. On Google Cloud, Pantheon Lab can choose to add GPUs to virtual machines (VMs) in Compute Engine or containers in [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (https://cloud.google.com/kubernetes-engine) (GKE).

Since the VMs of [Compute Engine](https://cloud.google.com/compute) (https://cloud.google.com/compute) can be customized and highly available, Pantheon Lab is able to create stable environments to test its products, which often requires non-stop computing for one to two days. By running the T4 GPU model on GKE through container instances, the company has also enhanced its development efficiency by reaching 25 times faster AI inferencing than using CPU machines.

"Computing performance is key to the success of our research and development work. Cloud GPUs provide us with industry-leading GPU models that can be flexibly deployed to meet our needs for ML model training, which leads to higher efficiency and smoother development process," notes Lau. "We can always have sufficient powerful computing resources whenever we want."

Storing sensitive data cost-effectively and securely with Cloud Storage

Offering cutting-edge technology means that Pantheon Lab needs to pay special attention to data security to protect its business secrets. By storing its ML model-related data in [Cloud Storage](https://cloud.google.com/storage) (https://cloud.google.com/storage), the company can

rest assured that its data is safeguarded by the high-level compliance standards that Google Cloud satisfies. On top of that, to store the same amount of data, the cost of Cloud Storage is more than eight times lower than solid-state drive (SSD) persistent disks, which is particularly beneficial for an AI company like Pantheon Lab that currently possesses a total of 25TB of valuable data.

"Besides computing power, data security is also one of our top priorities, because we have a lot of sensitive data," says Lau. "Cloud Storage offers trustworthy storage at a lower cost, enabling us to securely conduct our development work in a public cloud environment."

Optimizing digital humans and broadening the application

Pantheon Lab is now developing a software development kit (SDK) of its digital human solution using the computing resources of Google Cloud. With the SDK, external developers can leverage Pantheon Lab's technology to build different applications, which can further popularize the use of digital humans. The company also plans to explore ways to leverage PaLM 2, the latest large language model of Google, to develop more humanized chatbots.

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Lau says, "From technology advancement to product promotion, there is still a lot of potential for us to explore in the domain of digital humans. With Google Cloud facilitating our deep-learning research and development work, we're confident that we'll continue to bring better services to the market."